Experiment 6

Steady Two-Dimensional Heat Conduction in a Plate

Aim

To compare point-Jacobi iteration with a line-TDMA/SOR approach for the two-dimensional steady Laplace equation and assess convergence on two grids.

Simulation Parameters

Parameter	Value
Fine grid	81 × 61
Coarse grid	30 × 15
Heated-boundary temperature	100
Maximum iterations	10,000
Convergence tolerance	1 × 10■■

Numerical Method

The steady Laplace equation is discretised with second-order finite differences. The numerical field is obtained using point-Jacobi iteration and a line-implicit TDMA/SOR method, then compared with an analytical centre-line profile.

Boundary Conditions

The left boundary is prescribed as T(0,y) = T■ sin(πy/H). The right, top, and bottom boundaries are maintained at zero temperature.

Observations

Iteration counts, final residuals, relative L■ errors, centre-line profiles, grid resolution, and the influence of the relaxation factor were considered.

Results

Metric	Fine grid 81 × 61	Coarse grid 30 × 15
Jacobi iterations shown	10,000	~1,800
Line-TDMA iterations to tolerance	~800–900	~200
Jacobi final L∞ residual	~2 × 10■■	~10■■
Line-TDMA final L∞ residual	Below 10■■	Below 10■■
Centre-line profile agreement	Very good	Very good
Relative convergence speed	Slower	Faster

Conclusion

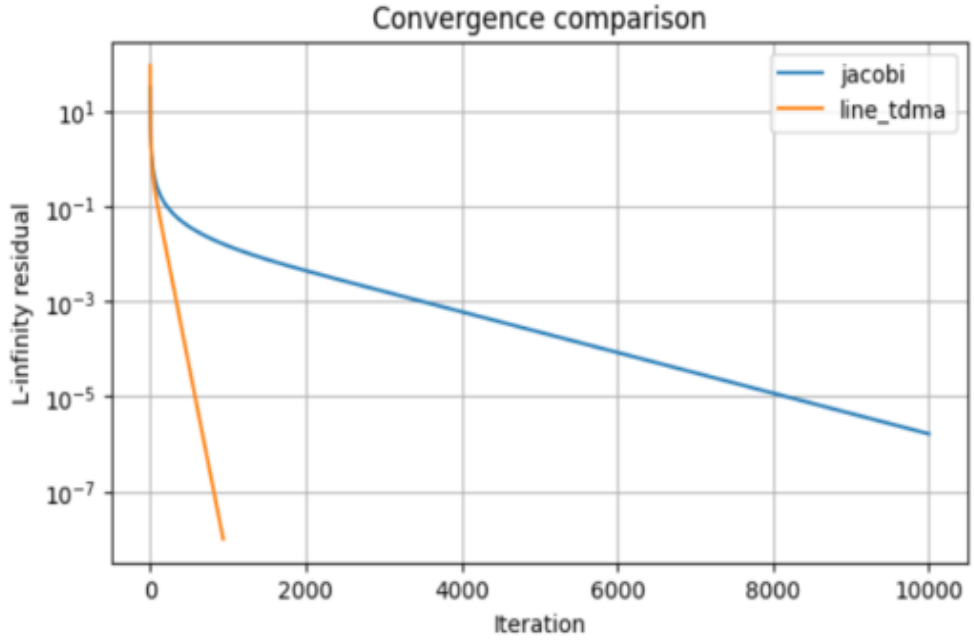
Both iterative approaches reproduce the expected centre-line temperature trend. Line-TDMA converges substantially faster than point-Jacobi in the reported cases. On the fine grid, Jacobi does not reach the specified tolerance within 10,000 iterations, whereas line-TDMA reaches it in roughly 800–900 iterations. The difference is especially clear on the coarse grid, where line-TDMA needs about 200 iterations.

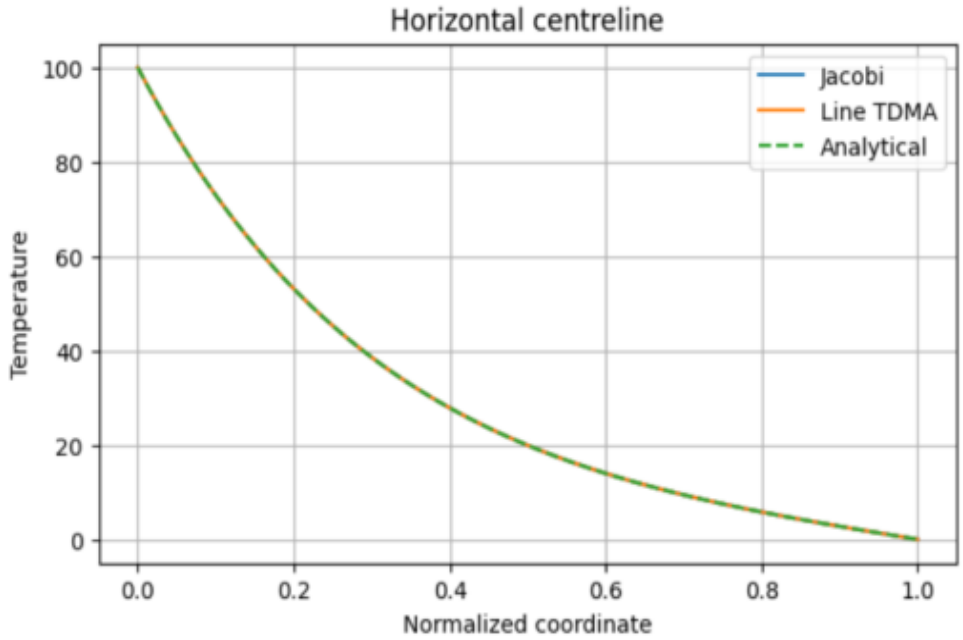
Original Simulation Figures

The following pages contain the plots and simulation outputs from the supplied report.

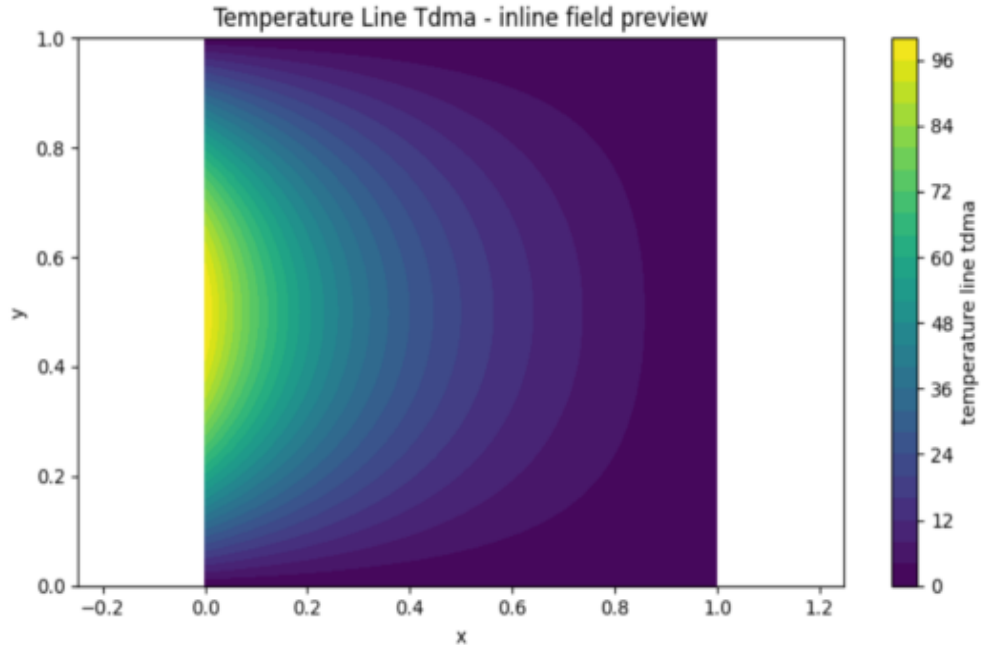
Original simulation output (source page 2)

For nx=81 and ny=61



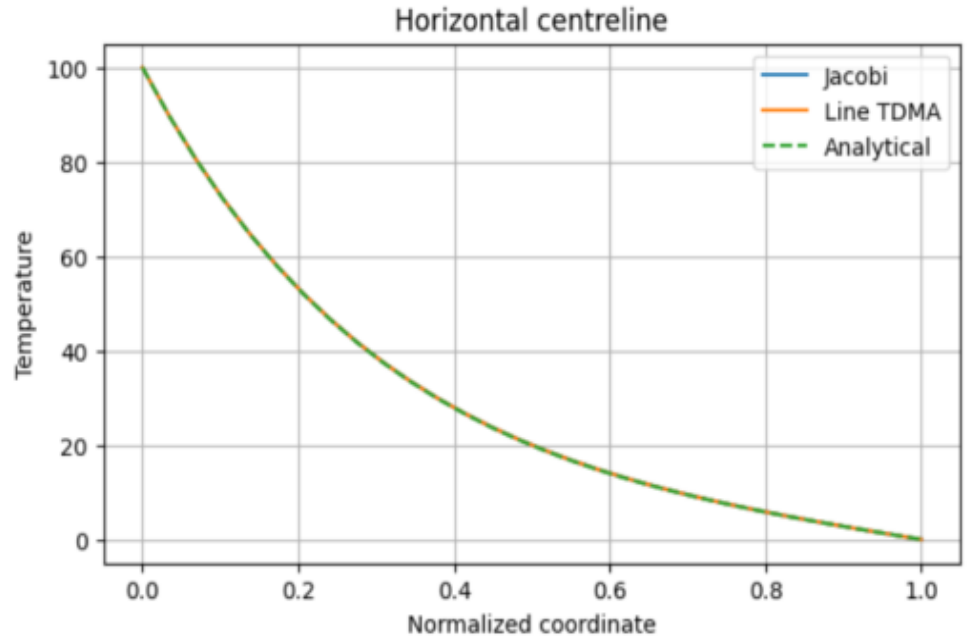


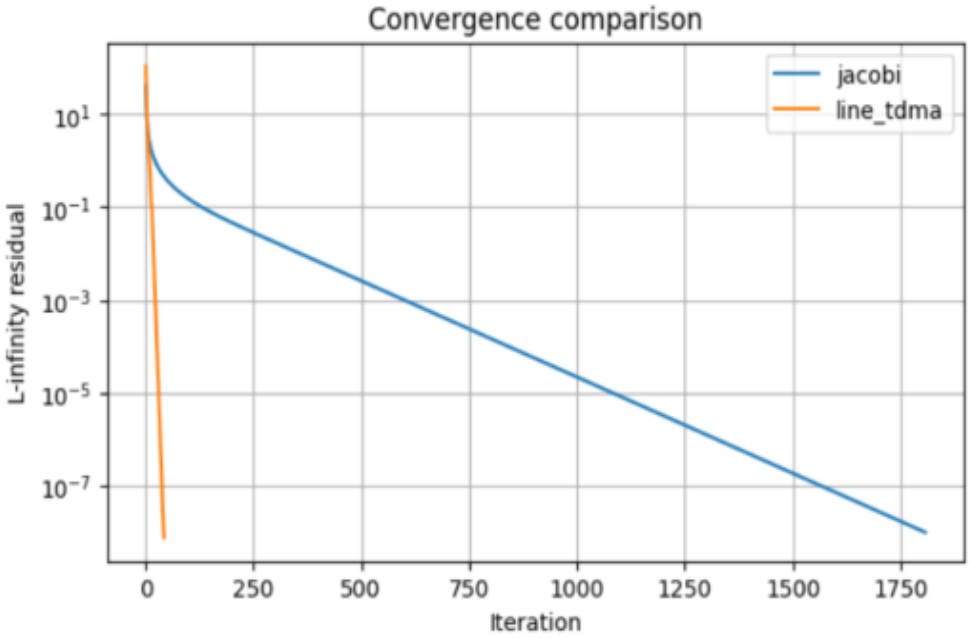
Original simulation output (source page 3)



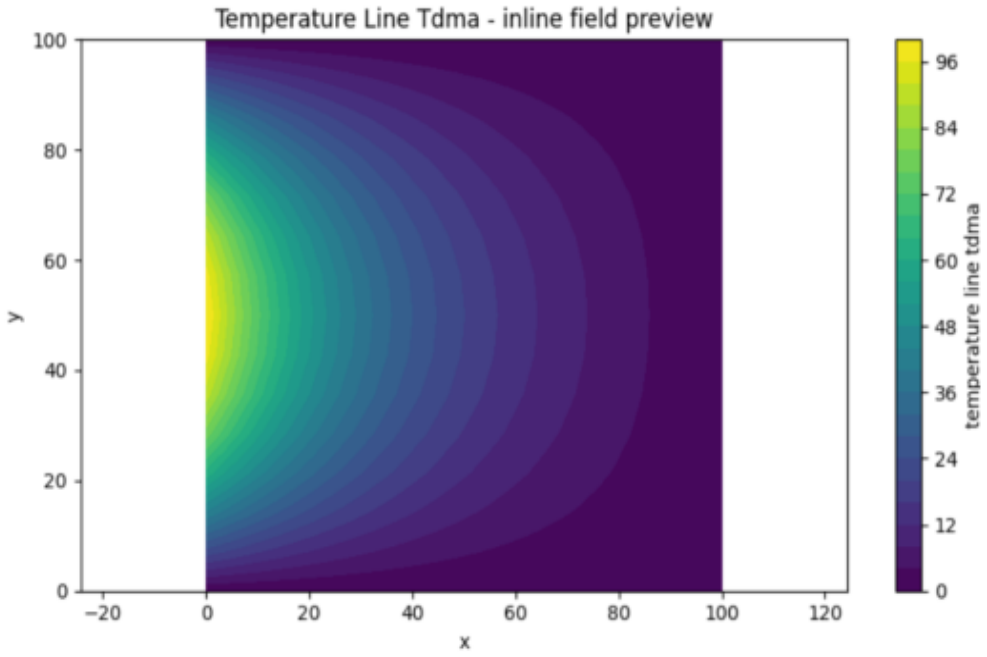
For nx=30 and ny=15

Original simulation output (source page 4)





Original simulation output (source page 5)



Results:

Parameter	**Fine Grid**	**Coarse Grid**
Grid size, N_x x N_y	81x61	30x15
Source temperature	100	100
Maximum iterations allowed	10,000	10,000
Convergence tolerance	10^{-9}	10^{-9}
Jacobi iterations shown	10,000	1,800
Line-TDMA iterations to reach tolerance	800–900	200

Original simulation output (source page 6)

Jacobi final L_{infty} residual	$2x10^{-6}$	10^{-9}
Line-TDMA final L_{infty} residual	Below 10^{-9}	Below 10^{-9}
Center line profile agreement	Very good	Very good
Relative convergence speed	Slower	Faster

Conclusion:

The two-dimensional steady heat conduction problem was solved using the Jacobi and line-TDMA iterative methods on grids of 81x61 and 30x15. Both methods produced temperature distributions that closely matched the analytical center line solution, with a maximum temperature of approximately 100 at the midpoint of the heated boundary and a gradual decrease toward the zero-temperature boundaries.

The convergence results showed that line-TDMA was substantially more efficient than Jacobi. On the coarse grid, Jacobi required approximately 1,800 iterations to reach the specified tolerance of 10^{-9}, whereas line-TDMA required approximately 200 iterations. On the fine grid, line-TDMA reached the tolerance within approximately 800–900 iterations, while Jacobi remained at a residual of approximately $2x10^{-6}$ after 10,000 iterations.

The results demonstrate that increasing grid resolution increases the computational effort required for convergence, especially for point-Jacobi iteration. The line-TDMA method handles this increase more efficiently because of its line-implicit solution procedure. Therefore, line-TDMA is the preferred method for this problem when faster convergence and satisfaction of a strict residual tolerance are required.